Asymmetric Distribution of Regional Medical Resources and its Policy Implications: Evidences from Jiangxi Province, China, 2023

Shiqi CHENG, Y3, School of Economics, University of Nottingham in Ningbo China

Introduction

The fair distribution of medical resources is essential for public health, but inequality remains a global challenge, especially in developing countries like China. Regional economic disparities further aggravate inefficient resource utilization and health inequality.

This study focuses on Jiangxi Province, where significant economic and medical disparities exist. In 2023, Nanchang, the provincial capital, accounted for 22.4% of the province's GDP, while less developed areas like Ganzhou lagged significantly behind. Paradoxically, medical resources are inversely distributed, with economically developed Nanchang having only 12.39% of hospitals versus 22.96% in less affluent Ganzhou, suggesting non-economic factors dominate this imbalance.

Research Questions

1. Does the utilization of medical resources in hospitals have an impact on the number of hospitalizations in Jiangxi Province?
2. Does the asymmetry of regional healthcare, based on the resource allocation related to urban economic levels (GDP), have an impact on the number of hospitalizations in Jiangxi Province?
3. Implications for policy in short-term and long-term

Data

The data was sourced from the Performance Analysis Report on DRGs in Jiangxi Province for the year 2023. The dataset contains medical record homepage information of 5,642,834 hospitalized patients from 331 hospitals.

Methodology

$$\ln Hospitalization = \beta_0 + \beta_1 DRG + \beta_2 AveD + \beta_3 AveC + \beta_4 \ln GDP + \mu_{(1)}$$

$$DRG = \alpha_0 + \alpha_1 \text{level} + \varepsilon_{(2)}$$

Level serves as the instrumental variable (IV)

Variable	Distribution
DGR	Total Disease Grouping Refinement (DRG): Relative Weight Sum per Case
AveD	Average Length of Stay in Hospital
AveC	Average cost of Stay in Hospital
ln_GDP	The logarithm of Gross Domestic Product in one City of Jiangxi Province
Level	Hospital grade, encoded as a categorical variable where 0 represents second-class hospitals, 1 represents tertiary hospitals

Empirical Results

Each hospital as one observation

Regression Results: OLS and IV Models

	(1) OLS_model	(2) IV_model
ln_GDP	-0.337**	-0.344*
	(0.111)	(0.153)
Average_cost	0.0000359*	-0.0000838
	(0.0000162)	(0.0000451)
DRG	0.0000245***	0.0000601***
	(0.00000204)	(0.0000120)
Average_duration	-0.0115**	0.0109
	(0.00417)	(0.00934)
Observations	331	331

Note: OLS and IV models for comparison.
First stage F-statistic: 25.82972068484097
"* p<0.05, ** p<0.01, *** p<0.001"

Results and Implications

- DRG can positively affect ln(hospitalization).
- ln(GDP) can negatively affect ln(hospitalization) reflects the unique characteristics of Jiangxi Province.
- In the short-term, government prioritize investments in developing areas to enhance local hospitals' capacity.
- In the long term, improving the tiered healthcare system and optimizing resource allocation across hospital levels can help balance accessibility and quality, fostering a more equitable healthcare system in Jiangxi Province.

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